
Bounded Precision-Geometry Scaling for Robust Multi-Task Learning under Loss Scale Mismatch

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Abstract

1 Multi-task learning often combines objectives with different numerical scales,
2 which can destabilize uncertainty-based loss weighting. We propose Bounded
3 Precision-Geometry Scaling (BPGS), a split uncertainty-weighting method that
4 maps each task’s latent uncertainty coordinate through a bounded sigmoid chart
5 and, in its batch-aware form, expresses task log-variances in detached batch log-loss
6 coordinates with first-batch calibration. We evaluate BPGS on synthetic stress tests,
7 NYUv2 dense prediction, Yeast multi-label classification, and RF1 multi-target
8 regression. Under pure post-hoc loss rescaling, BPGS keeps macro score nearly
9 unchanged from $\times 1$ to $\times 1000$ (0.777 to 0.778), whereas Kendall-style uncertainty
10 weighting drops from 0.780 to 0.637. On NYUv2, BPGS attains the best depth
11 absolute relative error (0.223), the best depth RMSE (0.790), and the lowest total
12 loss (1.891) among the compared methods. In an NYUv2 ablation, the batch-aware
13 variants avoid the severe degradation of the stateless auto-calibrated variant, and
14 the canonical batch-aware form improves depth and surface-normal metrics over
15 Kendall. On Yeast and RF1, BPGS remains competitive beyond synthetic stress
16 settings, achieving the best Yeast micro-F1 (0.616) and RF1 RMSE and MAE
17 within 0.562 and 0.450 of the leading baseline. These results support BPGS as
18 a robust uncertainty-weighting method when loss-scale mismatch is a primary
19 concern.

20 1 Introduction

21 Multi-task learning optimizes several objectives with shared parameters [1]. In practice, those
22 objectives often differ not only in semantics but also in numerical scale, curvature, and noise. This
23 mismatch appears in dense prediction, where segmentation, depth, and surface-normal losses coexist,
24 and in tabular settings that combine multiple binary or regression targets. When the weighting rule is
25 sensitive to raw loss magnitude, optimization can become brittle: one task can dominate the shared
26 update, and adaptive uncertainty variables can drift toward uninformative regimes.

27 Classical homoscedastic uncertainty weighting adapts task weights by learning one log-variance
28 parameter per task [2]. That formulation is effective in many settings, but its uncertainty coordinates
29 are unconstrained, so severe loss-scale mismatch can translate directly into large precision shifts.
30 Methods such as PCGrad [3] address a different failure mode by modifying conflicting gradients
31 rather than stabilizing the scale of learned uncertainty weights. This leaves a practical gap for a
32 method whose primary goal is robustness to heterogeneous task scales without giving up adaptive
33 weighting.

34 We study Bounded Precision-Geometry Scaling (BPGS), a bounded uncertainty-weighting method
35 designed for that gap. BPGS assigns one learnable uncertainty coordinate to each task, maps it
36 through a bounded sigmoid chart, and in its canonical form converts that bounded latent coordinate

into a batch-aware log-variance using detached log-loss statistics. The resulting precisions are used in a split optimization rule: network parameters are updated with detached normalized weights, while uncertainty coordinates are updated in a separate uncertainty objective. We do not position BPGS as a universal replacement for all multi-task optimizers. The question studied here is narrower: whether bounded, batch-aware uncertainty parameterization improves robustness under scale mismatch while remaining competitive on standard benchmarks.

Our contributions are empirical and methodological:

1. We formulate BPGS as a bounded, batch-aware, split uncertainty-weighting method for multi-task learning.
2. On a pure post-hoc loss-rescaling stress test, BPGS remains effectively invariant from $\times 1$ to $\times 1000$ scaling (macro score 0.777 to 0.778), whereas Kendall-style uncertainty weighting drops from 0.780 to 0.637.
3. On the main NYUv2 benchmark, BPGS attains the best depth absolute relative error (0.223), the best depth RMSE (0.790), and the lowest total loss (1.891) among the compared methods. A separate NYUv2 ablation shows that the batch-aware BPGS variants avoid the failure mode of the stateless auto-calibrated variant.
4. On Yeast multi-label classification and RF1 multi-target regression, BPGS remains competitive outside synthetic stress settings, achieving the highest Yeast micro-F1 (0.616) and RF1 RMSE and MAE within 0.562 and 0.450 of the leading baseline.

2 Related Work

Multi-task learning is commonly framed as shared-representation learning across related objectives [1]. Within that setting, adaptive weighting methods try to stabilize optimization by changing the scalar contribution of each task. The closest prior work to BPGS is homoscedastic uncertainty weighting, which learns one log-variance per task and uses that variable both to rescale the task loss and to regularize the uncertainty estimate [2]. Other adaptive weighting rules reweight tasks according to training-rate or performance signals, including GradNorm [4] and dynamic task prioritization [5]. BPGS is closest in spirit to the uncertainty-based line because it retains a separate uncertainty objective, but it changes the parameterization: the learnable coordinate is mapped through a bounded chart and then expressed in detached batch log-loss coordinates. The point of departure is geometric and scale-stability focused rather than a new gradient-balancing heuristic.

Another line of work treats multi-task learning as a multi-objective optimization problem and modifies gradients directly. Sener and Koltun derive a tractable surrogate for Pareto-style multi-objective optimization in deep multi-task learning [6]. PCGrad addresses harmful gradient interference by projecting conflicting task gradients [3], and CAGrad further regularizes the update toward conflict-averse directions [7]. These methods target a different failure mode from the one studied here. They are designed to manage competition in parameter space, whereas BPGS is designed to prevent uncertainty weights from becoming numerically brittle when raw task scales differ sharply. For that reason, PCGrad is an informative comparison baseline in our experiments, but it is not the closest conceptual predecessor.

3 Method

3.1 Problem setup

Let T denote the number of tasks, let w denote the shared network parameters, and let

$$L(w) = (L_1(w), \dots, L_T(w)) \quad (1)$$

collect the raw task losses for the current batch. The goal is to construct adaptive task weights that remain informative when the task losses differ substantially in scale. BPGS does this by learning one latent uncertainty coordinate per task and mapping that coordinate into a bounded log-variance chart.

82 3.2 Canonical BPGS chart

83 For the canonical batch-aware variant, BPGS computes detached log-loss statistics

$$\tilde{L}_i = \max(L_i, \varepsilon_{\log}), \quad \mu(L) = \frac{1}{T} \sum_{i=1}^T \log \tilde{L}_i, \quad (2)$$

84 and

$$\bar{\varsigma}(L) = \max \left(\sqrt{\frac{1}{T} \sum_{i=1}^T (\log \tilde{L}_i - \mu(L))^2}, \varepsilon_{\text{std}} \right). \quad (3)$$

85 Each task has a learnable uncertainty coordinate $\theta_i \in \mathbb{R}$. Using the task-count-dependent latent radius

$$\tau_T = \sqrt{T-1} + 0.1, \quad (4)$$

86 we define the bounded latent coordinate

$$z_i(\theta_i) = \tau_T (2\sigma(\theta_i) - 1), \quad (5)$$

87 where $\sigma(\cdot)$ is the logistic sigmoid. The canonical BPGS log-variance is then

$$s_i(\theta_i; L) = \mu(L) + \bar{\varsigma}(L) z_i(\theta_i). \quad (6)$$

88 The corresponding task precision is

$$\omega_i(\theta_i; L) = e^{-s_i(\theta_i; L)}, \quad (7)$$

89 and the network weight is its L1-normalized version

$$\alpha_i(\theta_i; L) = \frac{\omega_i(\theta_i; L)}{\sum_{j=1}^T \omega_j(\theta_j; L)}. \quad (8)$$

90 In the canonical configuration used for the main experiments, BPGS performs a one-shot auto-
 91 calibration step on the first observed batch. This initialization maps the first batch’s detached log-loss
 92 statistics into the bounded chart before the first network-weight construction. Ablation variants with
 93 fixed initialization or a stateless chart are studied separately, but the primary method studied in this
 94 paper is the batch-aware auto-calibrated form above.

95 3.3 Split optimization objectives

96 BPGS uses separate objectives for the network parameters and the uncertainty coordinates. The
 97 network objective is

$$J_{\text{net}}(w \mid \theta, L) = \sum_{i=1}^T \text{sg}[\alpha_i(\theta_i; L)] L_i, \quad (9)$$

98 where $\text{sg}[\cdot]$ denotes stop-gradient. The uncertainty objective is

$$J_{\text{unc}}(\theta \mid L) = \sum_{i=1}^T \left[\frac{1}{2} \omega_i(\theta_i; L) \text{sg}[L_i] + \frac{1}{2} s_i(\theta_i; L) \right]. \quad (10)$$

99 This matches the scalar uncertainty objective used by classical homoscedastic uncertainty weighting
 100 [2], but changes the chart from the learnable coordinate θ_i to the task log-variance s_i .

101 At each iteration, BPGS uses the same current-batch losses for both passes. The network parameters
 102 are updated with J_{net} , and the uncertainty coordinates are updated in a separate pass with J_{unc} . In
 103 the implemented method, the network pass uses detached normalized weights, while the uncertainty
 104 pass treats both raw losses and batch statistics as detached quantities.

105 3.4 Batch-conditional boundedness

106 The key formal property used in our robustness framing is conditional boundedness with respect to
 107 the current batch. For any fixed batch L and any finite θ_i ,

$$\mu(L) - \bar{\varsigma}(L)\tau_T < s_i(\theta_i; L) < \mu(L) + \bar{\varsigma}(L)\tau_T. \quad (11)$$

108 This follows immediately from $\sigma(\theta_i) \in (0, 1)$, which implies

$$z_i(\theta_i) \in (-\tau_T, \tau_T). \quad (12)$$

109 Substituting this interval into the affine map defining s_i gives the stated bound. Consequently, for any
 110 fixed batch, the induced precision $\omega_i(\theta_i; L)$ is strictly positive and remains inside a finite interval. The
 111 claim is therefore batch-conditional rather than globally independent of the observed loss statistics.

112 4 Experimental Setup

113 4.1 Benchmarks

114 We evaluate BPGS on five benchmark settings that serve different roles in the paper. First, we
 115 use the standard NYUv2 multi-task benchmark [8] for the main dense-prediction comparison. The
 116 benchmark contains 795 training images and 654 validation images, with three tasks: semantic
 117 segmentation, depth estimation, and surface-normal prediction. The main NYUv2 comparison uses a
 118 120-epoch budget and reports final-epoch metrics aggregated over seeds.

119 Second, we run a separate NYUv2 ablation study on a fixed 50% training subset while keeping the
 120 validation split unchanged. This study isolates the role of the bounded chart and the initialization rule
 121 by comparing four BPGS variants against Kendall uncertainty weighting: the canonical batch-aware
 122 auto-calibrated form, a batch-aware fixed-initialization form, a stateless fixed-initialization form, and
 123 a stateless auto-calibrated form. The ablation runs use a 60-epoch budget and the same three training
 124 seeds.

125 Third and fourth, we evaluate cross-regime generalization on two real-data multi-output problems.
 126 The Yeast benchmark [9] is a 14-label multi-label classification task with 1,500 training examples
 127 and 917 validation examples. The RF1 benchmark [10] is an 8-target regression task with 4,108
 128 training examples and 5,017 validation examples. In both cases, the tables report final task metrics
 129 aggregated over three seeds from runs scheduled for up to 120 epochs.

130 Finally, we use three synthetic robustness studies. The pure loss-rescaling study applies post-hoc scale
 131 multipliers $\times 1$, $\times 10$, $\times 100$, and $\times 1000$ to test invariance to raw loss magnitude. The scale-stress
 132 study uses the same scale grid in a synthetic multi-task setting designed to probe degradation under
 133 scale imbalance. The heterogeneous mixed-stress study evaluates three regimes, denoted *clean*,
 134 *conflict*, and *noisy*, to test robustness when tasks differ not only in scale but also in interaction
 135 structure.

136 4.2 Compared methods and metrics

137 The exact comparison set depends on the benchmark. The main NYUv2 benchmark compares
 138 BPGS against Static, Kendall uncertainty weighting [2], and UWSO. The NYUv2 ablation compares
 139 the four BPGS variants against Kendall. The Yeast and RF1 benchmarks compare BPGS against
 140 Static, Kendall, PCGrad [3], a GradNorm-inspired gradient-norm proxy [4], and UWSO. The pure
 141 loss-rescaling and scale-stress studies compare BPGS, Kendall, and UWSO, while the heterogeneous
 142 mixed-stress study compares BPGS, Kendall, PCGrad, and UWSO.

143 For NYUv2, we report segmentation mIoU, depth absolute relative error, depth RMSE, mean angular
 144 error for normals, normal accuracy within 11.25° , total validation loss, and in the main benchmark
 145 also Δ_M against the Static baseline [11]. For Yeast, we report macro-F1, micro-F1, and Hamming
 146 accuracy. For RF1, we report R^2 , RMSE, and MAE. For the synthetic stress studies, the primary
 147 statistic is macro score, with worst-task score used in the heterogeneous analysis and the stress
 148 diagnostics.

Table 1: Pure loss rescaling (06): Macro Score across scale factors. Mean $\pm$ std over 3 seeds.

Method	$\times 1$	$\times 10$	$\times 100$	$\times 1000$
Kendall	0.780 ± 0.007	0.769 ± 0.013	0.714 ± 0.017	0.637 ± 0.015
UWSO	0.622 ± 0.073	0.674 ± 0.036	0.688 ± 0.035	0.681 ± 0.014
BPGS	0.777 ± 0.012	0.784 ± 0.008	0.781 ± 0.001	0.778 ± 0.006

4.3 Reporting protocol

All tables aggregate over the same three training seeds: 42, 43, and 44. Unless stated otherwise, values are reported as mean $\pm$ standard deviation across seeds. For the synthetic stress studies, we aggregate per-seed summary statistics. For NYUv2, Yeast, and RF1, we report the final recorded epoch of each run. The code and configuration files needed to reproduce the experiments are available at <https://github.com/neryva-lab/spectra>.

One reporting rule is important for fairness on NYUv2. Checkpoint summaries are not uniformly available across methods, so mixing best-checkpoint and final-checkpoint numbers would introduce method-dependent extraction rules. We therefore compute all main NYUv2 results from final-epoch metrics for every method. The real-data tables are also reported from final task metrics, while the run configurations use `val/total_loss` as the training selection metric. This keeps the paper tables tied to a single extraction rule across methods and seeds.

5 Results

We organize the evaluation around four questions: whether BPGS is robust to explicit scale perturbations, whether that robustness transfers to a standard dense-prediction benchmark, which parts of the method matter in ablation, and whether the method remains competitive outside the synthetic stress setting.

5.1 Robustness to loss-scale mismatch

Figure 1 separates two perturbation types. The right column isolates pure post-hoc loss rescaling, where only the numerical scale changes. In that setting, BPGS is effectively invariant: its macro score stays at 0.777, 0.784, 0.781, and 0.778 from $\times 1$ through $\times 1000$ (Table 1 and Appendix Figure 3). Kendall-style uncertainty weighting starts slightly higher at $\times 1$ but drops monotonically to 0.637 at $\times 1000$, while UWSO improves modestly from a much weaker starting point.

The left column of Figure 1 is the harder test because the perturbation changes the optimization problem rather than only its units. All three methods degrade as the factor increases, but BPGS is best at $\times 1$, $\times 100$, and $\times 1000$, reaching 0.529 at $\times 1000$ versus 0.495 for Kendall and 0.486 for UWSO. The relative drop from $\times 1$ to $\times 1000$ is 29.1% for BPGS, compared with 33.3% for Kendall and 31.7% for UWSO (Appendix Table 6; see also Appendix Figure 2). In our experiments, this provides the strongest support for the bounded, batch-aware construction.

The heterogeneous mixed-stress study is less favorable to BPGS than the explicit scale-mismatch studies (Appendix Table 7 and Appendix Figure 5). Kendall and PCGrad attain higher macro scores in the noisy and conflict regimes, whereas BPGS gives the strongest worst-task score in the clean regime at 0.301 and remains far above the UWSO worst-task score of 0 throughout. We therefore interpret BPGS as a method targeted at loss-scale robustness rather than as a general solution to heterogeneous task interaction.

5.2 Main NYUv2 benchmark

Table 2 shows that no method dominates every metric. Static achieves the best segmentation mIoU, and UWSO is strongest on the two surface-normal metrics. BPGS, however, gives the strongest depth results and a strong aggregate trade-off by the reported summary metrics: it attains the best absolute relative depth error (0.223), the best depth RMSE (0.790), the lowest total loss (1.891), and the highest Δ_M (0.078). The same trade-off pattern is visible in Appendix Figures 8–11.

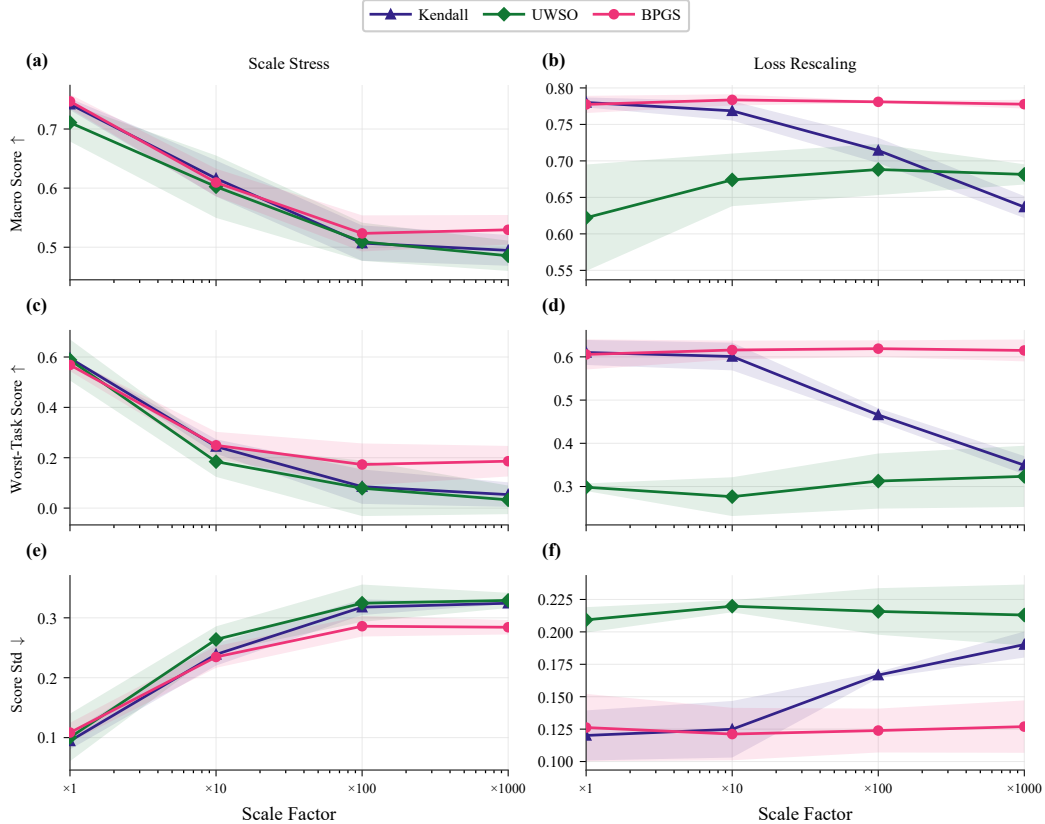


Figure 1: Synthetic robustness diagnostics under scale mismatch. Left column: scale stress, where performance is evaluated as task scales become increasingly imbalanced. Right column: pure loss rescaling, where only the reported magnitude changes. BPGS remains nearly invariant under pure rescaling and retains the strongest macro score at the largest scale-stress factors.

Table 2: NYUv2 dense prediction benchmark. All methods trained for 120 epochs. Final-epoch metrics reported as mean $\pm$ std over 3 seeds. Δ_M computed against Static baseline.

Method	mIoU $\uparrow$	Abs Rel $\downarrow$	RMSE $\downarrow$	Angle $\downarrow$	Within 11.25° $\uparrow$	Total Loss $\downarrow$	Δ_M $\uparrow$
Static	0.341 $\pm$ 0.006	0.239 $\pm$ 0.000	0.829 $\pm$ 0.011	35.934 $\pm$ 2.055	0.169 $\pm$ 0.009	1.992 $\pm$ 0.033	0.000 $\pm$ 0.000
Kendall	0.281 $\pm$ 0.013	0.229 $\pm$ 0.001	0.808 $\pm$ 0.003	28.629 $\pm$ 0.431	0.231 $\pm$ 0.007	1.977 $\pm$ 0.029	0.022 $\pm$ 0.000
UWSO	0.294 $\pm$ 0.007	0.228 $\pm$ 0.003	0.815 $\pm$ 0.012	26.110 $\pm$ 0.383	0.271 $\pm$ 0.007	1.936 $\pm$ 0.023	0.059 $\pm$ 0.000
BPGS	0.312 $\pm$ 0.001	0.223 $\pm$ 0.002	0.790 $\pm$ 0.010	26.851 $\pm$ 0.334	0.257 $\pm$ 0.005	1.891 $\pm$ 0.014	0.078 $\pm$ 0.000

Relative to the strongest competing values, BPGS improves absolute relative error from 0.228 under UWSO to 0.223 and RMSE from 0.808 under Kendall to 0.790, while also lowering total loss from 1.936 to 1.891. The NYUv2 result therefore supports a specific claim: the robustness-oriented BPGS parameterization does not have to trade away standard benchmark performance to control scale sensitivity.

5.3 Ablation on chart and initialization

Table 3 isolates the role of the batch-aware chart and the initialization rule. The stateless auto-calibrated variant degrades severely, with mIoU 0.058 and total loss 6.142. This shows that first-batch auto-calibration is not sufficient on its own; in these ablations, the batch-aware chart that re-expresses the uncertainty variables in current log-loss coordinates is necessary for stable behavior. Appendix Figure 6 shows this failure in the validation dynamics, while Appendix Figure 7 shows the contrasting task-weight behavior of the stable variants.

Table 3: Ablation study: BPGS variants vs. Kendall uncertainty weighting on NYUv2. All methods trained for 60 epochs. Mean $\pm$ std over 3 seeds.

Method	mIoU $\uparrow$	Abs Rel $\downarrow$	RMSE $\downarrow$	Angle $\downarrow$	Within 11.25° $\uparrow$	Total Loss $\downarrow$
Kendall	0.137 $\pm$ 0.006	0.294 $\pm$ 0.010	0.990 $\pm$ 0.016	38.681 $\pm$ 0.936	0.130 $\pm$ 0.005	2.546 $\pm$ 0.035
BPGS (canonical)	0.128 $\pm$ 0.008	0.283 $\pm$ 0.005	0.988 $\pm$ 0.043	33.834 $\pm$ 1.068	0.161 $\pm$ 0.015	2.488 $\pm$ 0.075
BPGS (BA-fixed)	0.143 $\pm$ 0.006	0.291 $\pm$ 0.005	0.994 $\pm$ 0.022	39.097 $\pm$ 0.478	0.130 $\pm$ 0.004	2.524 $\pm$ 0.047
BPGS (SL-auto)	0.058 $\pm$ 0.006	0.331 $\pm$ 0.003	1.305 $\pm$ 0.015	34.661 $\pm$ 1.310	0.147 $\pm$ 0.024	6.142 $\pm$ 1.332
BPGS (SL-fixed)	0.120 $\pm$ 0.007	<u>0.287 $\pm$ 0.006</u>	1.013 $\pm$ 0.025	35.183 $\pm$ 0.963	0.142 $\pm$ 0.010	2.566 $\pm$ 0.051

Table 4: Real-data transfer benchmarks. Final-epoch task metrics reported as mean $\pm$ std over 3 seeds. Higher is better for Yeast metrics and RF1 R^2 ; lower is better for RF1 RMSE and MAE.

Method	Macro-F1 $\uparrow$	Micro-F1 $\uparrow$	Hamming Acc $\uparrow$	R^2 $\uparrow$	RMSE $\downarrow$	MAE $\downarrow$
Static	0.430 $\pm$ 0.008	0.610 $\pm$ 0.009	0.772 $\pm$ 0.003	-0.453 $\pm$ 0.127	29.875 $\pm$ 0.485	23.376 $\pm$ 0.451
Kendall	<u>0.431 $\pm$ 0.003</u>	0.609 $\pm$ 0.001	0.768 $\pm$ 0.002	-0.439 $\pm$ 0.164	<u>29.801 $\pm$ 0.821</u>	<u>23.219 $\pm$ 0.790</u>
UWSO	0.142 $\pm$ 0.014	0.486 $\pm$ 0.008	0.767 $\pm$ 0.002	-0.408 $\pm$ 0.193	30.144 $\pm$ 1.137	23.449 $\pm$ 1.230
PCGrad	0.430 $\pm$ 0.007	0.604 $\pm$ 0.010	0.769 $\pm$ 0.002	-0.432 $\pm$ 0.107	29.596 $\pm$ 0.385	23.098 $\pm$ 0.368
GradNormProxy	0.408 $\pm$ 0.028	0.597 $\pm$ 0.014	<u>0.772 $\pm$ 0.001</u>	-0.508 $\pm$ 0.248	30.033 $\pm$ 0.482	23.380 $\pm$ 0.582
BPGS	0.436 $\pm$ 0.006	0.616 $\pm$ 0.008	0.773 $\pm$ 0.003	<u>-0.429 $\pm$ 0.213</u>	30.158 $\pm$ 0.952	23.548 $\pm$ 0.908

Among the stable variants, the best values split between the two batch-aware forms. The canonical batch-aware BPGS variant gives the best depth absolute relative error, the best depth RMSE, the best mean angular error, the best normal accuracy within 11.25°, and the lowest total loss. The batch-aware fixed-initialization variant gives the best mIoU. By contrast, the stateless fixed variant remains competitive on depth but does not match the batch-aware variants on the strongest metrics. The ablation therefore supports a narrow design conclusion: the batch-aware chart is the critical component, while the exact initialization rule mainly changes which task trade-off is favored among otherwise stable runs.

5.4 Transfer to Yeast and RF1

The real-data benchmarks are less separable than the synthetic stress tests, so we interpret them as evidence of competitiveness rather than universal dominance. On Yeast, BPGS is best on all three reported metrics, with macro-F1 0.436, micro-F1 0.616, and Hamming accuracy 0.773 (Table 4; Appendix Figure 12). On RF1, BPGS is not the top method: PCGrad gives the lowest RMSE and MAE, and UWSO gives the best mean R^2 . BPGS nonetheless remains close to the better RF1 baselines, with $R^2 = -0.429$, RMSE 30.158, and MAE 23.548 (Appendix Figure 13).

Taken together, these results bound the scope of the paper’s empirical claim. BPGS is strongest where loss-scale mismatch is the central difficulty, remains strong on NYUv2, and stays competitive on two additional real-data multi-output problems. The results do not support a stronger statement that BPGS is uniformly best across all benchmarks and metrics.

6 Discussion and Limitations

The empirical case for BPGS is strongest on the scale-mismatch questions the method was designed to address. The rescaling study isolates numerical-scale invariance directly, the scale-stress study tests degradation under worsening imbalance, and the NYUv2 results show that the same parameterization remains competitive on a standard dense-prediction benchmark. The paper therefore supports a targeted claim about robustness-oriented uncertainty weighting rather than a general claim of dominance over all multi-task optimizers.

The study also has clear limits. First, the benchmark set is small: one dense-prediction benchmark, two tabular real-data problems, and several controlled stress tests. Second, all headline results are aggregated over three seeds, which is enough to expose large failures such as the stateless auto-calibrated ablation, but not enough to settle fine-grained ranking differences when methods are close. Third, the paper does not establish convergence guarantees or scale-independent optimality claims; the boundedness result in Section 3 is batch-conditional and does not imply global optimization guarantees. Fourth, BPGS depends on current-batch log-loss statistics, so its behavior may change under different batch sizes or substantially different noise regimes from those tested here. Fifth,

although the method introduces only task-level scalar uncertainty updates, the paper does not provide a dedicated runtime or hardware-efficiency study.

Potential societal impact is indirect. A more stable task-weighting rule can improve the reliability of multi-output models used in scientific or engineering pipelines, but any downstream deployment still inherits the data, labeling, and evaluation failures of the underlying tasks. In safety-sensitive settings, a method that improves one task while weakening another can be misused if only aggregate performance is reported, which is why this paper reports per-task metrics and states where BPGS is not the best method.

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A Reproducibility Details

All main-paper tables aggregate over seeds 42, 43, and 44. The stress studies use per-seed `results.csv` summaries. NYUv2, Yeast, and RF1 tables are extracted from epoch-level `metrics.csv` logs using the final recorded epoch for each run. This final-epoch rule is especially important for NYUv2 because checkpoint metadata is not consistently available across methods.

Table 5: Synthetic scale stress (02): Macro Score across scale factors. Mean $\pm$ std over 3 seeds.

Method	$\times 1$	$\times 10$	$\times 100$	$\times 1000$
Kendall	0.742 ± 0.010	0.616 ± 0.030	0.507 ± 0.030	0.495 ± 0.026
UWSO	0.711 ± 0.032	0.602 ± 0.053	0.509 ± 0.032	0.486 ± 0.026
BPGS	0.747 ± 0.012	0.610 ± 0.023	0.523 ± 0.031	0.529 ± 0.025

Table 6: Relative Macro Score change (%) from $\times 1 \rightarrow \times 1000$. Positive values indicate degradation; negative values indicate improvement.

Method	Scale Stress	Loss Rescaling
Kendall	33.3%	18.4%
UWSO	31.7%	−9.5%
BPGS	29.1%	0.0%

Main NYUv2 benchmark. The standard NYUv2 benchmark uses 795 training images and 654 validation images with semantic segmentation, depth estimation, and surface-normal prediction. Representative BPGS runs use 120 epochs, batch size 8, learning rate 5×10^{-4} , minimum learning rate 10^{-6} , 100 warmup steps, weight decay 10^{-4} , gradient clipping 10.0, and mixed-precision training. The configuration files include early-stopping fields, but the paper tables are extracted from the final epoch for every method.

NYUv2 ablation. The ablation uses a fixed 50% training subset with the same validation split. Representative runs use 60 epochs, batch size 8, learning rate 5×10^{-4} , minimum learning rate 10^{-6} , 100 warmup steps, weight decay 10^{-4} , gradient clipping 1.0, and mixed precision.

Yeast and RF1. Yeast uses 1,500 training examples and 917 validation examples; RF1 uses 4,108 training examples and 5,017 validation examples. Representative Yeast BPGS runs use 120 epochs, batch size 128, learning rate 5×10^{-4} , minimum learning rate 10^{-5} , 200 warmup steps, weight decay 10^{-4} , gradient clipping 10.0, and selection metric `val/total_loss` with patience 15. Representative RF1 BPGS runs use the same optimizer settings, batch size 256, and the same selection rule.

Method naming. The real-data comparison includes GradNormProxy, which is a gradient-norm proxy baseline inspired by GradNorm rather than a claim of faithful reimplement. The UWSO baseline is an analytical inverse-loss weighting rule with fixed temperature as implemented in this codebase.

Submission-scope disclosures. The experiments use standard public benchmarks cited in the main text. An anonymized code repository with the full implementation, configuration files, and reproduction scripts is available at <https://github.com/neryva-lab/spectra>. Uniform hardware and wall-clock records for every archived run are not included, so the corresponding checklist item is answered conservatively.

B Additional Stress Results

C Ablation and NYUv2 Figures

D Real-Data Training Curves

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Table 7: Heterogeneous mixed stress (07): Macro Score and Worst-Task Score across regimes. Mean $\pm$ std over 3 seeds.

Method	Clean		Noisy		Conflict	
	Macro	Worst	Macro	Worst	Macro	Worst
Kendall	0.687 ± 0.010	0.267 ± 0.056	0.659 ± 0.007	0.242 ± 0.045	0.664 ± 0.012	0.211 ± 0.041
UWSO	0.437 ± 0.028	0.000 ± 0.000	0.438 ± 0.020	0.000 ± 0.000	0.433 ± 0.013	0.000 ± 0.000
PCGrad	0.688 ± 0.016	0.274 ± 0.055	0.656 ± 0.007	0.241 ± 0.046	0.660 ± 0.011	0.205 ± 0.048
BPGS	0.679 ± 0.017	0.301 ± 0.052	0.641 ± 0.013	0.176 ± 0.076	0.652 ± 0.012	0.198 ± 0.063

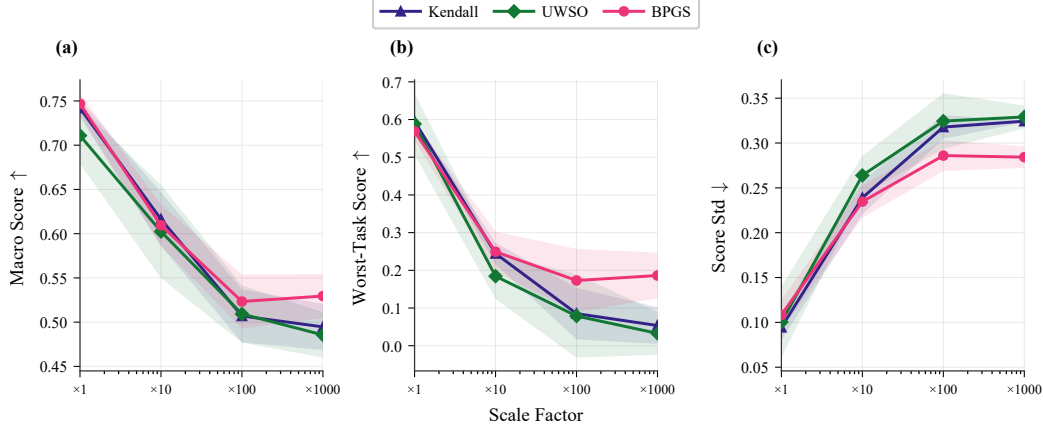


Figure 2: Synthetic scale-stress curves. As the scale factor increases, all methods degrade, but BPGS retains the strongest macro score at $\times 1$, $\times 100$, and $\times 1000$, and tracks the strongest worst-task score at the larger stress levels.

Answer: [Yes]

Justification: The abstract and introduction state the contribution as robustness under loss-scale mismatch and explicitly avoid a claim of universal superiority; see the abstract and Section 1.

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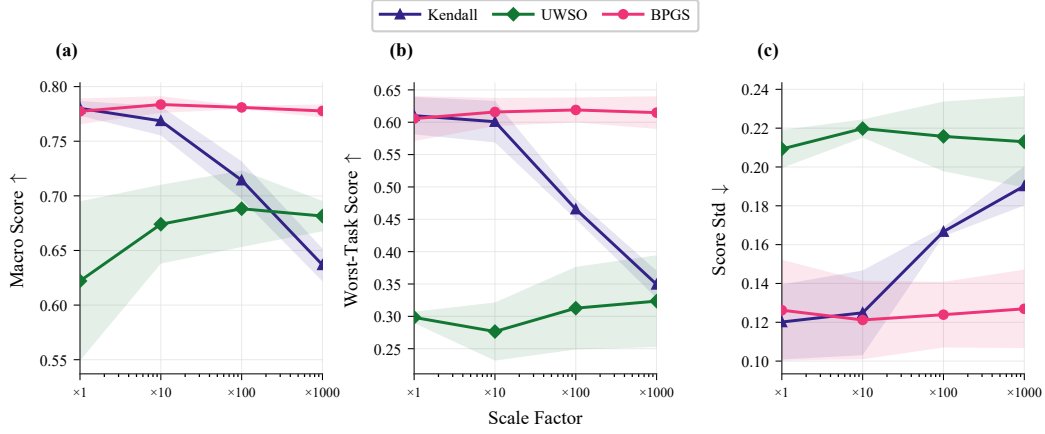


Figure 3: Pure loss-rescaling curves. BPGS remains nearly invariant across the entire rescaling range, while Kendall degrades sharply at larger factors and UWSO improves only from a substantially weaker starting point.

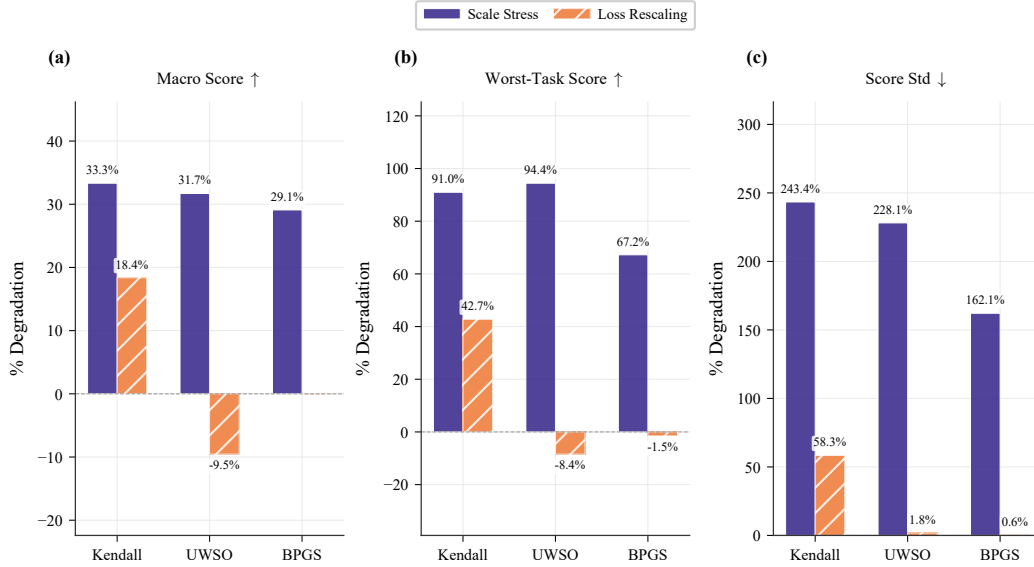


Figure 4: Relative degradation comparison between synthetic scale stress and pure loss rescaling. The figure shows that BPGS has the smallest macro-score degradation under scale stress and essentially no macro-score degradation under pure rescaling.

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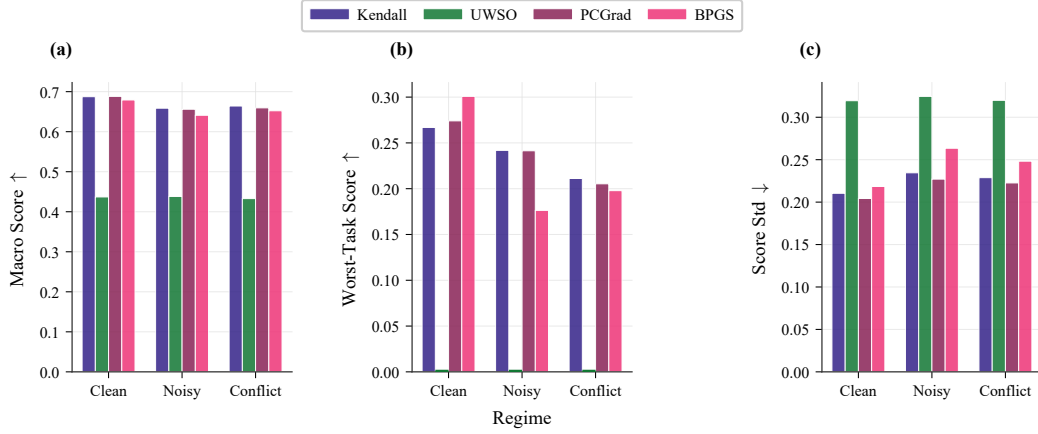


Figure 5: Heterogeneous mixed-stress summary. BPGS is not the best method in every regime, but it achieves the strongest worst-task score in the clean regime and remains well above the degenerate UWSO worst-task score across all regimes.

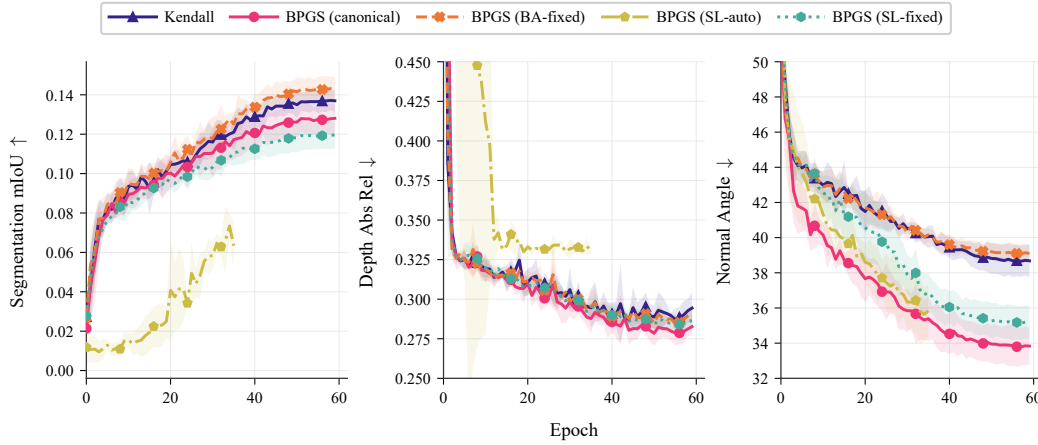


Figure 6: NYUv2 ablation validation curves by task. The stateless auto-calibrated variant fails on segmentation and remains weak on depth, while the batch-aware variants maintain stronger trajectories throughout training.

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Justification: Section 3.4 states the batch-conditional boundedness result, its assumptions, and the complete proof argument based on the bounded sigmoid chart.

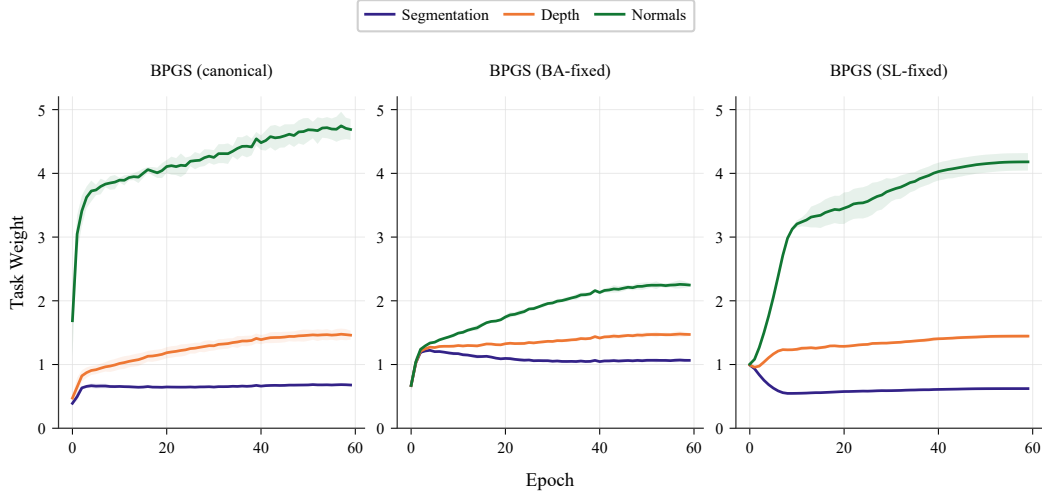


Figure 7: Task-weight dynamics for representative stable BPGS variants on the NYUv2 ablation. The batch-aware variants keep task weights in distinct but controlled regimes, whereas the stateless fixed variant shows a different long-run allocation pattern, especially for the surface-normal task.

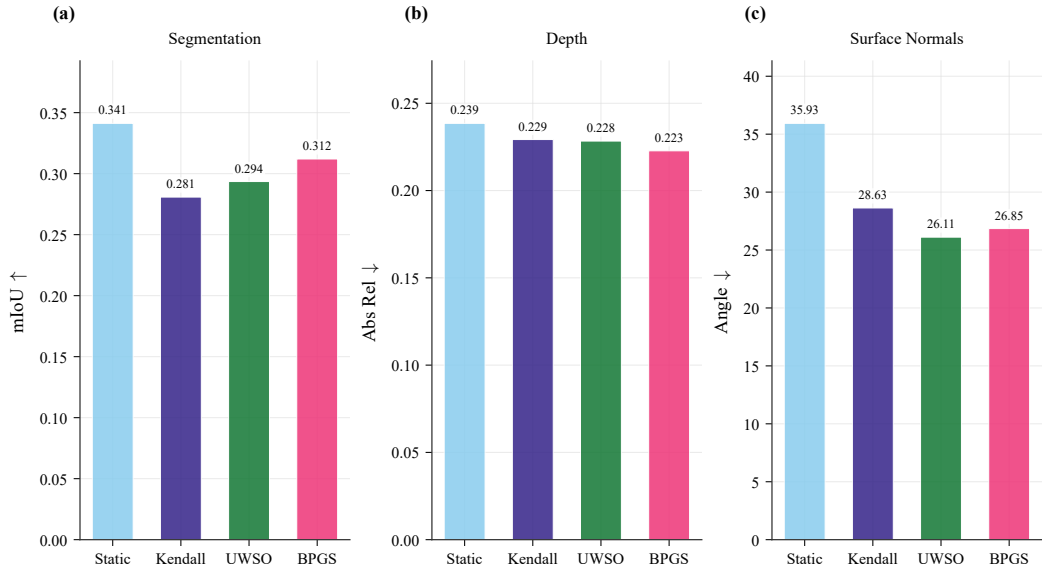


Figure 8: NYUv2 per-task performance comparison. The figure makes the trade-off pattern visually explicit: Static is strongest on segmentation, UWSO is strongest on surface normals, and BPGS is strongest on both depth metrics.

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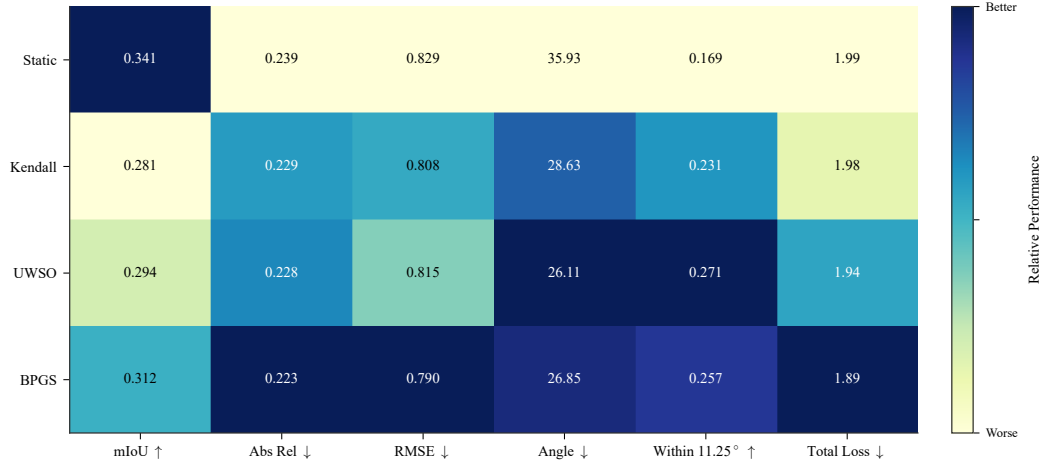


Figure 9: NYUv2 performance heatmap across reported metrics. Darker cells indicate stronger relative performance after accounting for metric direction. BPGS concentrates its gains on depth and total loss, while Static and UWSO remain stronger on some other tasks.

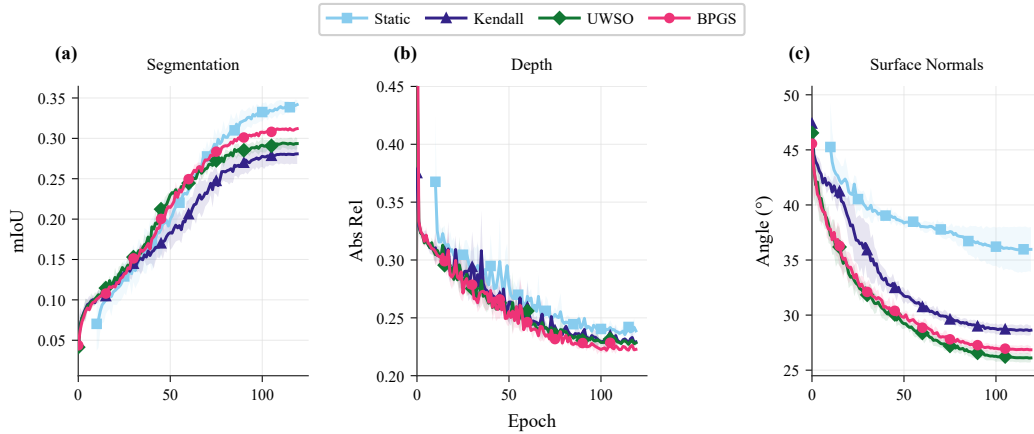


Figure 10: NYUv2 validation training curves by task. BPGS tracks the strongest depth trajectory and remains competitive on segmentation and surface normals over the full 120-epoch budget.

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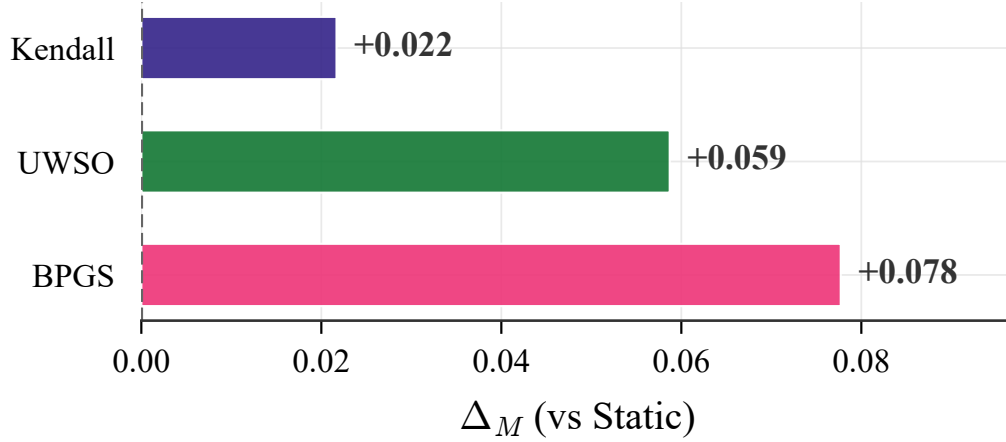


Figure 11: NYUv2 Δ_M comparison against the Static baseline. BPGS attains the strongest aggregate improvement among the adaptive methods evaluated in the main benchmark.

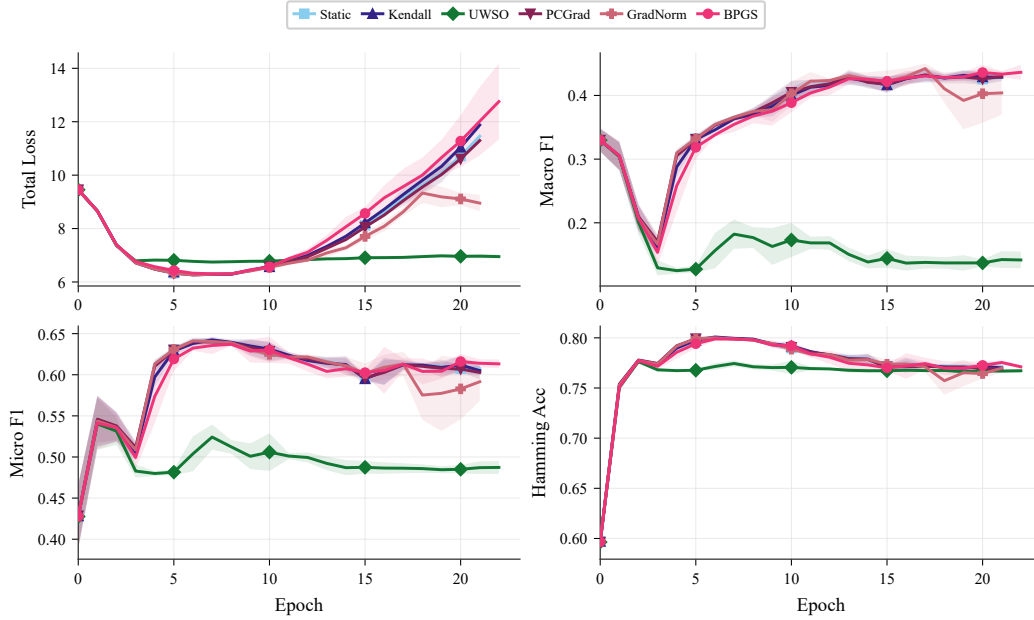


Figure 12: Yeast training curves. BPGS tracks the strongest final micro-F1 and macro-F1 among the compared methods, while UWSO remains substantially weaker throughout training.

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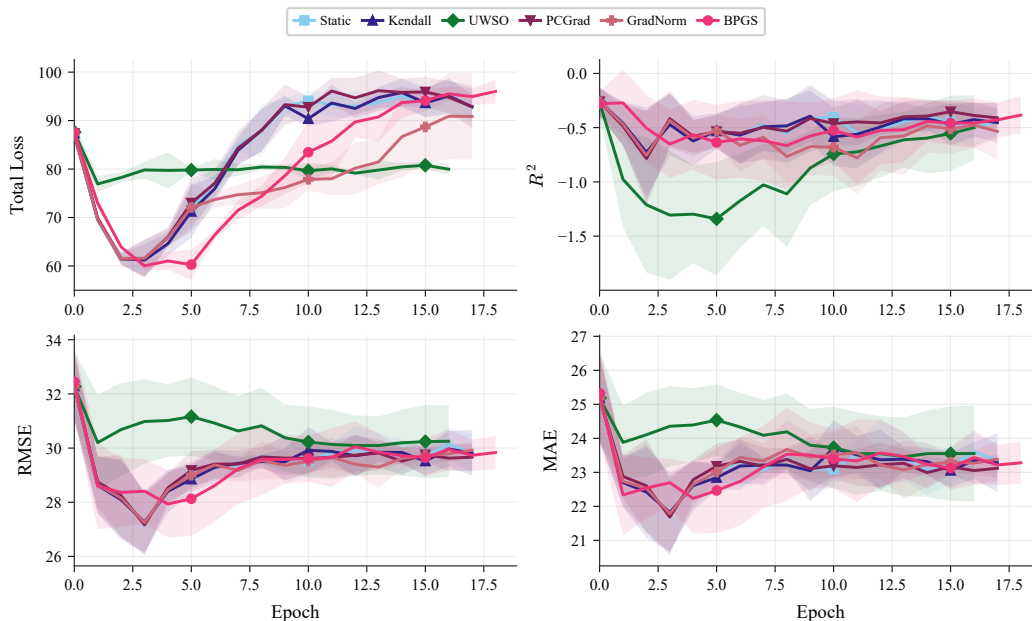


Figure 13: RF1 training curves. The methods remain closer than in the synthetic stress studies, which is consistent with the paper’s claim that RF1 supports competitiveness rather than clear superiority.

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